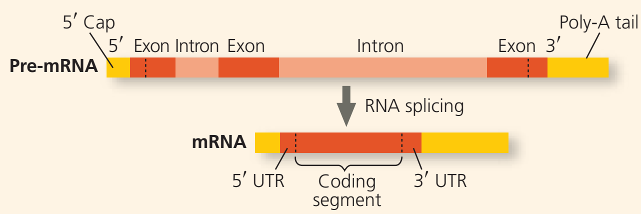
some cases, RNA alone catalyzes its own splicing. The properties of RNA allow some RNAs (called **ribozymes**) to act as catalysts. The presence of introns allows for **alternative RNA splicing**.

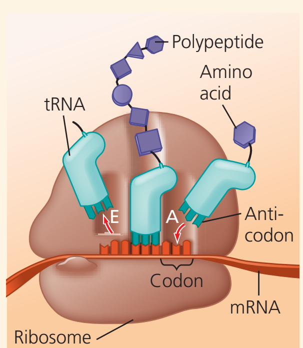


What function do the 5′ cap and the poly-A tail serve on a eukaryotic mRNA?

CONCEPT 17.4

Translation is the RNA-directed synthesis of a polypeptide: *A Closer Look* (pp. 347–356)

- A cell translates an mRNA message into protein using **transfer RNAs (tRNAs)**. After being bound to a specific amino acid by an **aminoacyl-tRNA synthetase**, a tRNA lines up via its **anticodon** at the complementary codon on mRNA. A **ribosome**, made up of **ribosomal RNAs (rRNAs)** and proteins, facilitates this coupling with binding sites for mRNA and tRNA.
- Ribosomes coordinate the three stages of translation: initiation, elongation, and termination. The formation of peptide bonds between amino acids is catalyzed by rRNAs as tRNAs move through the **A** and **P sites** and exit through the **E site**.



- After translation, during protein processing, proteins may be modified by cleavage or by attachment of sugars, lipids, phosphates, or other chemical groups.
- Free ribosomes in the cytosol initiate synthesis of all proteins, but proteins with a **signal peptide** are synthesized on the ER.
- A gene can be transcribed by multiple RNA polymerases simultaneously. Also, a single mRNA molecule can be translated simultaneously by a number of ribosomes, forming a **polyribosome**. In bacteria, these processes are coupled, but in eukaryotes they are separated in space and time by the nuclear membrane.

How do tRNAs function in the context of the ribosome during translation?

CONCEPT 17.5

Mutations of one or a few nucleotides can affect protein structure and function (pp. 357–362)

- Small-scale **mutations** include **point mutations**, changes in one DNA nucleotide pair, which may lead to production of nonfunctional proteins. **Nucleotide-pair substitutions** can cause **missense** or **nonsense mutations**. Nucleotide-pair **insertions** or **deletions** may produce **frameshift mutations**.
- Spontaneous mutations can occur during DNA replication, recombination, or repair. Chemical and physical **mutagens** cause DNA damage that can alter genes.
- The **CRISPR-Cas9 system** is a powerful new **gene-editing** technique with the potential to correct genetic mutations that cause disease. Significant technical and ethical questions have been raised about how this could or should be used.

What will be the results of chemically modifying one nucleotide base of a gene? What role is played by DNA repair systems in the cell?

TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. In eukaryotic cells, transcription cannot begin until
(A) the two DNA strands have completely separated and exposed the promoter.
(B) several transcription factors have bound to the promoter.
(C) the 5′ caps are removed from the mRNA.
(D) the DNA introns are removed from the template.

2. Which of the following is true of a codon?
(A) It never codes for the same amino acid as another codon.
(B) It can code for more than one amino acid.
(C) It can be either in DNA or in RNA
(D) It is the basic unit of protein structure.

3. The anticodon of a particular tRNA molecule is
(A) complementary to the corresponding mRNA codon.
(B) complementary to the corresponding triplet in rRNA.
(C) the part of tRNA that bonds to a specific amino acid.
(D) catalytic, making the tRNA a ribozyme.

4. Which of the following is true of RNA processing?
(A) Exons are cut out before mRNA leaves the nucleus.
(B) Nucleotides are added at both ends of the RNA.
(C) Ribozymes may function in the addition of a 5′ cap.
(D) RNA splicing adds a poly-A tail to the mRNA.

5. Which component is directly involved in translation?
(A) RNA polymerase
(B) ribosome
(C) spliceosome
(D) DNA

Levels 3-4: Applying/Analyzing

6. Using Figure 17.6, identify a 5′ → 3′ sequence of nucleotides in the DNA template strand for an mRNA coding for the polypeptide sequence Phe-Pro-Lys.
(A) 5′-UUUCCCAAA-3′
(B) 5′-GAACCCCTT-3′
(C) 5′-CTTCGGGAA-3′
(D) 5′-AAACCCUUU-3′

7. Which of the following mutations would be *most* likely to have a harmful effect on an organism?
(A) a deletion of three nucleotides near the middle of a gene
(B) a single nucleotide deletion in the middle of an intron
(C) a single nucleotide deletion near the end of the coding sequence
(D) a single nucleotide insertion downstream of, and close to, the start of the coding sequence